Software Requirements Specification (SRS) Template

Project: File Conversion Service(API)
Version: 1.0
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Revision history

Version	Date	Author	Change summary	Approval
1.0	18-08-2025	Anivartha u Ananya Joshi Ayush Gowda	PES	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	ARVIND SRINATH K T		22-08-2025

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1. Introduction

1.1 Purpose

This document specifies the requirements for a **File Conversion Service API** that allows users and applications to convert files from one format to another (e.g., Markdown → PDF, CSV → JSON, DOCX → TXT). It defines functional and non-functional requirements, system features, external interfaces, and acceptance criteria.

1.2 Scope

The system will:

- Provide an API for developers and end-users to upload a file and request conversion into a supported format.
- Support multiple conversion types (text, document, spreadsheet, markup, etc.).
- Return the converted file through download or API response.
- Handle errors gracefully (e.g., unsupported format, corrupted file).

1.3 Audience

Developers, End Users, QA Engineers, System Administrators.

1.4 Definitions

API,CSV,PDF,JSON,MD(Light-Weight markup language)

2. Overall description

2.1 Product perspective

This is a standalone web-based API service. It will interact with:

- File upload/download endpoints
- Internal conversion engine (libraries/tools)
- Logging and monitoring services

2.2 Major product functions (detailed)

- Accept file upload in supported formats
- Convert file to requested target format
- Return converted file
- Provide metadata about conversions (size, time, type)
- Handle errors (unsupported format, corrupted file)

2.3 User roles and characteristics (expanded)

- **API Users (Developers):** Basic programming knowledge, integrate API with apps.
- **End-Users (optional Web UI):** Upload file, choose target format, download result.
- **Admin:** Monitor service, view logs, manage formats supported.

2.4 Operating environment

- Backend: Node.js + Express + TypeScript
- Database: MySQL (for logs, metadata)
- Deployment: Cloud / On-prem
- Protocols: REST over HTTPS

2.5 Constraints

- Limited to supported file formats.
- Maximum file size configurable (e.g., 50 MB).
- Requires internet connection (if deployed as SaaS).

3. External interface requirements

3.1 User interfaces

- Web-based upload interface (optional): Simple UI to upload and download files.
- API Documentation (Swagger/Postman): For developers.

3.2 Hardware interfaces

No specialized hardware required. Standard web server hosting sufficient.

3.3 Software interfaces

- Conversion libraries (e.g., Pandoc for documents, Python/Node packages for formats).
- Logging & Monitoring (ELK stack / Prometheus).

3.4 Communications

- REST API over HTTPS (TLS 1.2+).
- JSON-based request/response format for API.

4. Functional - features(detailed)

Each requirement below includes acceptance criteria and a reference test case.

Name	Req ID	Requirement	Type	Priority	Acceptance Criteria
File Upload &Validation	FCS-F-01	System shall allow users to upload files in supported formats	Functional	High	Upload succeeds if file type and size valid.
File Conversion	FCS-F-02	System shall convert uploaded file into selected target format	Functional	High	Converted file retains readable content
Downlaod/ Response Delivery	FCS-F-03	System shall allow users to download converted files or receive response.	Functional	High	Download file availabewithout file corruption
Error Handling	FCS-F-04	System shall provide descriptive error messages for unsupported formats, corrupted files, or oversized files.	Functional	High	Error messages are clear, accurate, and guide user to resolution.
Logging & Monitoring	FCS-F-05	Log all requests with timestamps, file types, and status	Functional	high	Generate logs for each request with correct details

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
FCS-NF-001	95% conversions < 3s for files ≤ 10MB	Performance	High	Measured via load testing

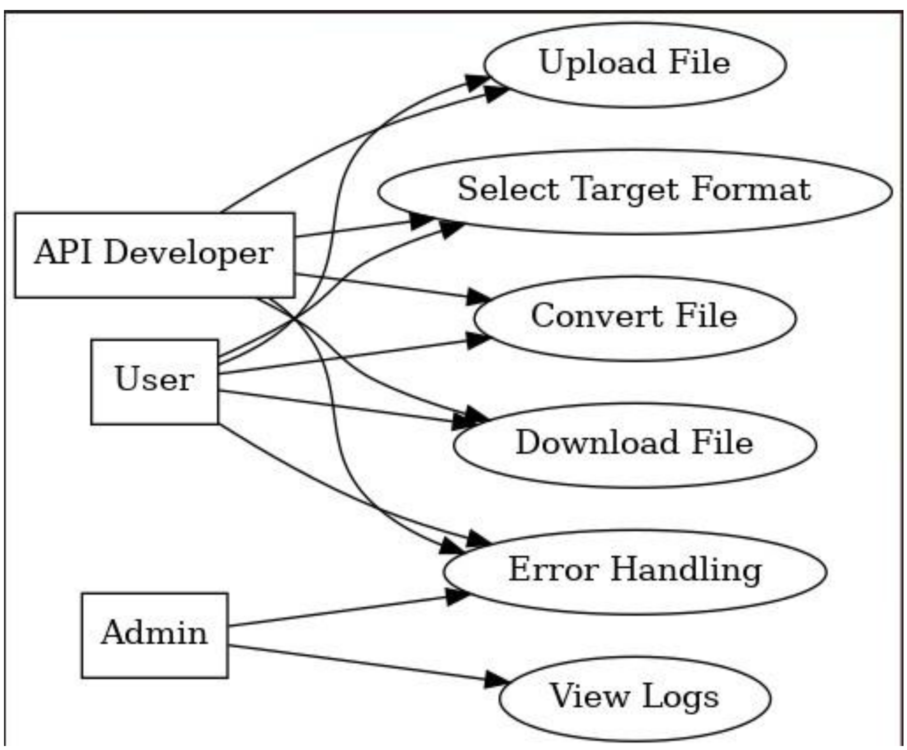
FCS-NF-002	99.5% uptime monthly	Reliability	High	Monitoring logs
FCS-NF-003	TLS 1.2+ encryption mandatory	Security	High	Pen-test verified
FCS-NF-004	Logs retained for 1 year	Audit	Medium	Log archival verified
FCS-NF-005	API documentation auto-generated	Usability/Accessibility	Medium	Swagger UI available

6. Quality attributes & Acceptance tests

- **Acceptance Criteria:** All high-priority requirements implemented and verified.
- **Test Suites:** Upload tests, Conversion tests, Error handling tests, Performance tests, Security tests.

7. System models and diagrams

7.1 UML Use-Case diagram



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
FCS-F-001	Upload File	4.1	UploadModule	TC-UP-01	N	
FCS-F-002	Convert File	4.2	ConversionEngine	TC-CV-01	N	
FCS-F-003	Download Response	4.3	DeliveryModule	TC-DL-01	N	
FCS-F-004	Error Handling	4.4	ErrorHandler	TC-ER-01	N	
FCS-NF-001	Performance < 3s	5	PerfMonitor	TC-Perf-01	N	